

RICK BHATTACHARYA

MECHANICAL ENGINEERING STUDENT | UNIVERSITY OF NORTH CAROLINA AT CHARLOTTE (GPA: 3.7)

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Full Interactive Portfolio: <https://rickbhattacharya.vercel.app/>

I'm a mechanical engineering student at UNC Charlotte who got into rocketry by wanting to understand how things actually work — not just simulate them. Most of my projects started because I couldn't find an off-the-shelf solution that made sense, so I built one. I'm looking for internship or co-op opportunities where I can contribute real engineering work, whether that's in propulsion, mechanical design, mechatronics, or something else entirely. I learn fast and I work well in environments where the problems are hard.

TECHNICAL PROFICIENCIES

CAD & DESIGN	ANALYSIS & SOFTWARE	FABRICATION & HANDS-ON
SolidWorks, Fusion 360, AutoCAD, Onshape, Siemens NX, Creo Parametric, GD&T, DFM	STAR-CCM+, NASA CEA, Rocket Propulsion Analysis (RPA), Microsoft Visio (P&ID), OpenRocket, MATLAB, Python, ANSYS	3D Printing, Machining-ready Drawings, Prototype Assembly, Fit Checks, Rapid Iteration, Test Setup, Soldering, KiCAD

SELECTED ENGINEERING PROJECTS

ROLE: LEAD ENGINEER & DESIGNER

Designing a 20 lbf bipropellant liquid rocket engine on a \$500 budget. The constraint is real — every component decision has to be justified. The injector uses an impinging orifice geometry, sized using Microsoft Excel to model choked flow conditions and confirm the orifices would stay sonic at our target mass flow rates. Currently in the works of building the injector and going to be running a flow simulation on OpenFOAM or STAR-CCM+ to verify choked flow.

Used Microsoft Excel to solve isentropic flow equations and size the injector throat areas for choked flow — the goal was keeping the discharge coefficient within a 2% margin of the theoretical value across the orifice geometry.

Ran flow simulations to check bipropellant mixing efficiency and pressure drop across the injector. The numbers gave me more confidence in the design before committing to hardware.

Built out full P&ID diagrams for the feed system. Using COTS components kept cost down while still allowing safe, controllable regulation of high-pressure propellants.

KEY RESULTS

- Full design architecture built under \$500 through deliberate COTS selection.
- Excel choked-flow solvers used to size orifices within 2% of theoretical.
- P&ID and flow analysis completed before any physical parts were ordered.

ROLE: TECHNICAL LEAD — CAROLINA LIQUID ROCKETRY

Led the design study for a 250 lbf bipropellant liquid rocket engine through Carolina Liquid Rocketry. The program reached a complete parametric CAD model and a full-scale 3D printed mockup before being paused pending funding for hardware. The design uses a pintle injector architecture — a geometry I studied through Halfcat's open-source work and adapted for our propellant combination and chamber sizing.

Used NASA CEA to determine propellant mixture ratios, chamber pressure, and specific impulse targets. Those numbers drove every downstream geometry decision in the design.

Built a fully parametric CAD model in Fusion 360 — when performance requirements shifted, I could propagate changes across all five subsystem interfaces without rebuilding anything from scratch.

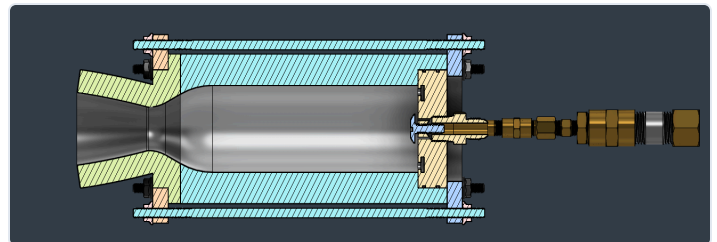
Before the program paused, I printed a full-scale mockup to do physical fit checks. It caught three interference issues that would have been expensive to find after machining. The metal hardware design is ready; the program is waiting on funding to move forward.

KEY RESULTS

- Complete parametric CAD model across five subsystems — ready for manufacturing when the program resumes.
- Three assembly conflicts identified through physical mockup before any machined parts were ordered.
- NASA CEA analysis established propellant ratios, chamber pressure, and Isp baseline for the full design.



FULL-SCALE 3D PRINTED MOCKUP



ENGINE CAD — CROSS SECTION



ENGINE 3D RENDER

ROLE: INDEPENDENT PROJECT

Built an active Thrust Vector Control system for a high-power rocket — a 2-axis gimbal that uses real-time IMU data to drive servo corrections and keep the rocket pointed where it's supposed to go. The hardware went through seven distinct CAD revisions before I was satisfied with the printability and load paths.

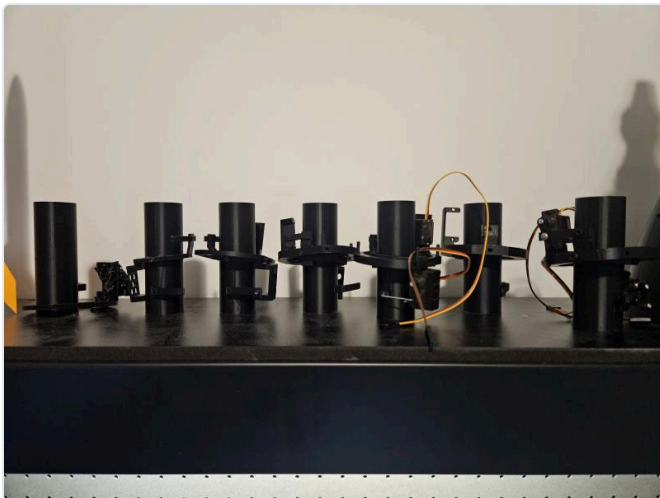
Seven CAD iterations on the motor mount and servo linkages — each one driven by a specific printability or structural problem from the previous version. I applied DFM principles throughout so the printed parts would actually hold up under thrust loads.

Wrote PID firmware for closed-loop attitude control. The controller pulls gyroscope and accelerometer data at high frequency and drives servo corrections fast enough to counter aerodynamic disturbances.

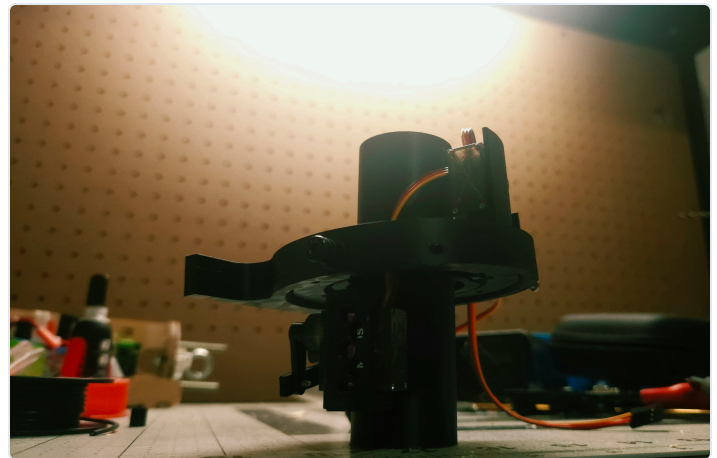
Built a ground test bench to validate the gimbal's range of motion before any flight attempt. Confirmed $\pm 15^\circ$ pitch and yaw authority with the response times the control loop needed.

KEY RESULTS

- Seven CAD revisions driven by DFM — each one a specific fix, not just a refresh.
- $\pm 15^\circ$ actuation range confirmed on a hardware test bench before committing to flight.
- Closed-loop PID firmware written from scratch for onboard attitude control.



GIMBAL TEST VIDEO



TVC GIMBAL CAD ITERATIONS



FUNCTIONAL GIMBAL PROTOTYPE

ROLE: INDEPENDENT PROJECT

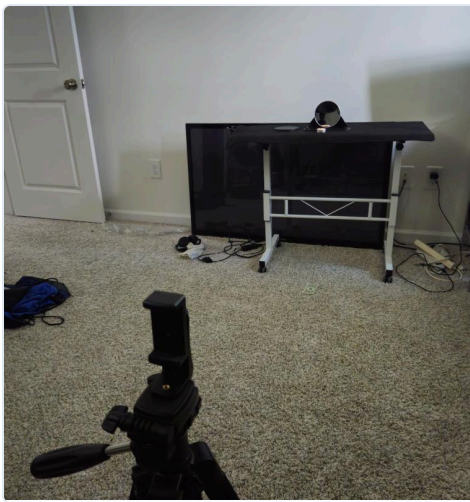
Built a benchtop Schlieren imaging system from scratch to actually see airflow — thermal plumes, density gradients, the kind of stuff you normally need a wind tunnel budget to observe. The whole point was having a real physical check on CFD assumptions without needing expensive lab access.

Machined and assembled custom mirror mounts to hold precise optical alignment through the light path. Small vibrations ruin the image, so the mount stability mattered a lot.

The system works by refracting light through air density gradients — you can see things like heat rising off a hand or airflow separating around an object. It's a useful sanity check on flow field assumptions that would otherwise only exist in simulation.

KEY RESULTS

- Thermal plumes and air currents successfully visualized and recorded.
- Gives a physical reference point for comparing against CFD flow field results.



OPTICAL TEST SETUP



MIRROR ALIGNMENT

ROLE: INDEPENDENT PROJECT

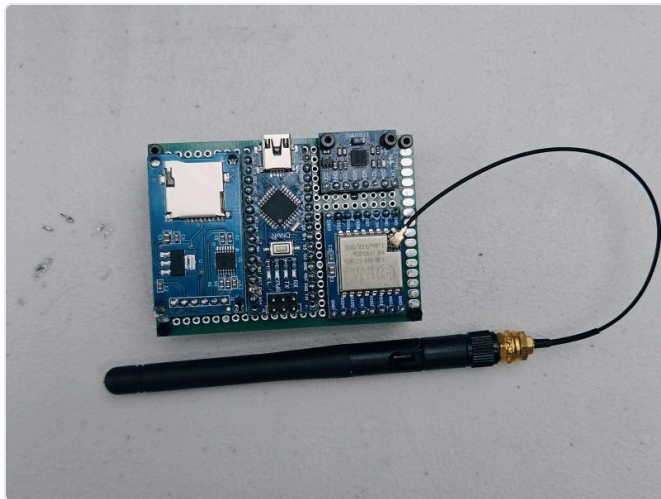
Designed and built a small-form-factor flight computer because I wanted to know exactly what was happening during a flight — not just read back whatever a commercial altimeter decided to save. The board integrates barometric and inertial sensors and logs data at a rate that actually captures the interesting parts of the flight profile.

Laid out the schematic and PCB in KiCAD, keeping mass and footprint tight enough to fit inside standard airframe tubes. Hand-soldered all the SMD components myself.

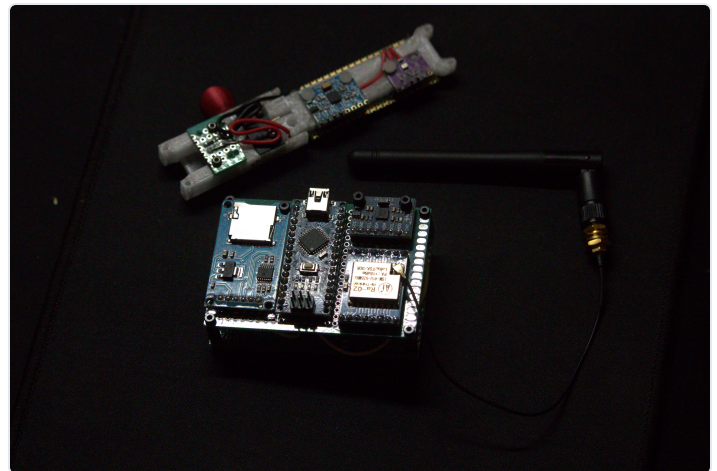
Wrote embedded C/C++ firmware to handle sensor polling and wireless telemetry. Getting the timing right for high-frequency data logging while also managing the radio link was the most involved part of the firmware work.

KEY RESULTS

- Real-time altitude, velocity, and orientation data logged and transmitted during flight.
- Full-stack build — schematic, layout, assembly, and firmware all done independently.



ASSEMBLED SMD FLIGHT COMPUTER



MINI FLIGHT COMPUTER LAYOUT

ROLE: INDEPENDENT PROJECT

Building a high-power rocket to earn my Tripoli L1 and L2 certifications. The avionics bay uses a pull-pin arming system — a mechanical interlock that keeps the ejection charges electrically isolated until the pin is physically pulled at the pad. It's a simple, reliable safety mechanism that removes any ambiguity about whether the system is armed.

The AV bay houses dual-deploy altimeters, GPS, and battery management in a compact package. The layout is designed so everything is accessible during pad prep without needing to partially disassemble the rocket.

The pull-pin interlock is straightforward by design: until the pin comes out, the e-matches have no continuity path. No software states to worry about, no way to accidentally arm it.

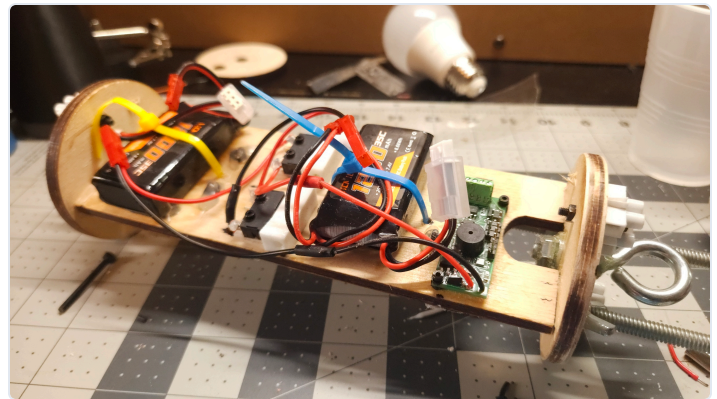
This vehicle is also a test platform for the custom avionics hardware I've been developing — the flight data will feed directly into improving the next iteration of the flight computer.

KEY RESULTS

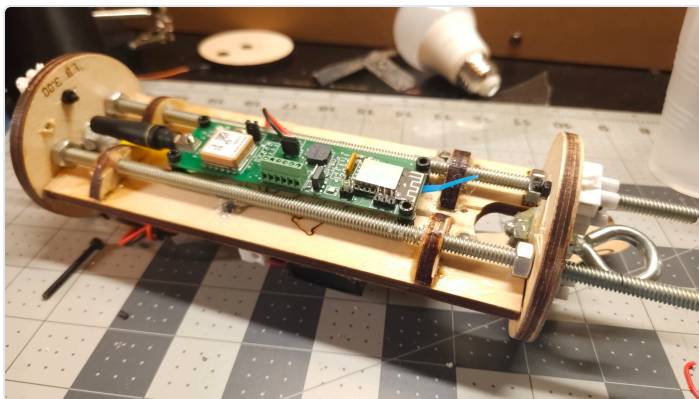
- Pull-pin interlock gives unambiguous physical confirmation the system is safe on the pad.
- Dual-deploy system for independent main and drogue deployment.
- Live flight test platform for custom avionics hardware development.



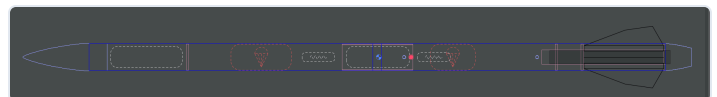
VEHICLE — ASSEMBLY



AV BAY — FRONT (PULL-PIN ARMING)



AV BAY — BACK (ELECTRONICS)



OPENROCKET SIMULATION

EXPERIENCE

Worked with the solid propulsion group at ICSP in Kolkata, contributing to their research rocket program. My work was primarily analytical — nozzle geometry design, propellant grain calculations, and pulling together the performance specs for the motor configurations they were studying.

Worked through propellant grain geometry calculations to understand how burn surface area evolves during a firing and how that translates into thrust over time. The geometry decisions have a direct effect on the shape of the thrust curve, so getting those numbers right matters.

Designed the nozzle geometry for the motor — sizing throat area and expansion ratio based on the target chamber pressure and desired exit conditions. Put together the full rocket specs to document the design and give the team a clear reference for the configuration.